2D Finite Element Analysis of a Microstrip Transmission Line

A verified numerical evidence chain for capacitance and impedance

Group C

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MTH321 Numerical PDEs — Project 2

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Problem and engineering goal

What we need

- Compute the quasistatic capacitance per unit length.
- Reuse the same mesh for the all-air capacitance.
- Convert the matched pair into $\varepsilon_{\mathrm{eff}}$ and Z_0.

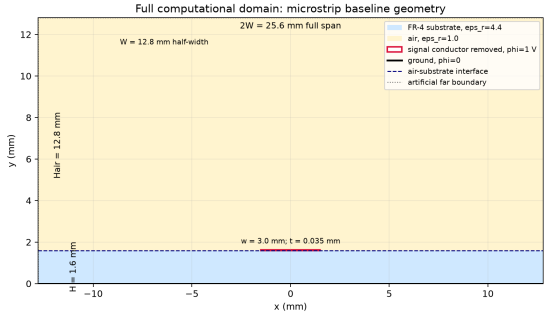
What would not be enough

- A single FEM number near $50\,\Omega$.
- Agreement with an empirical formula only.
- Unchecked corner and boundary effects.

Takeaway

The result matters because the numerical evidence chain makes it defensible.

Geometry and PDE model



Electrostatic model

$$-\nabla \cdot (\varepsilon \nabla \phi) = 0$$

$$\phi = 1\,\mathrm{V} \text{ on trace}, \qquad \phi = 0 \text{ on ground}$$

Homogeneous Neumann conditions are used on the top and side artificial boundaries.

Physical dimensions

$H = 1.6\,\mathrm{mm}$, $w = 3.0\,\mathrm{mm}$, $t = 0.035\,\mathrm{mm}$, substrate $\varepsilon_r = 4.4$.

Takeaway

A finite 8H domain approximates the open cross-section.

Weak form and P1 FEM

Weak problem

Find $\phi \in V_g$ such that $\int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega = 0$ for all $v \in \mathcal{V}$.

Discretisation

- Conforming triangular P_1 elements.
- Trace interior excluded; Dirichlet DOFs eliminated.
- Positive permittivity gives an SPD free system.

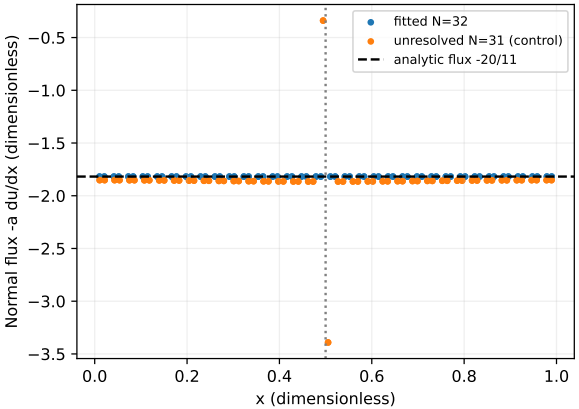
Material interface

Potential is continuous; flux balance holds in the **weak sense**. Discrete normal fluxes need not match pointwise on each edge.

Takeaway

The weak form handles the material jump; stability and convergence rate are different questions.

Solver Verification: Smooth and Layered Problems



Layered interface

Analytical: dashed black
Fitted: agrees near roundoff
Unresolved: flux error at interface

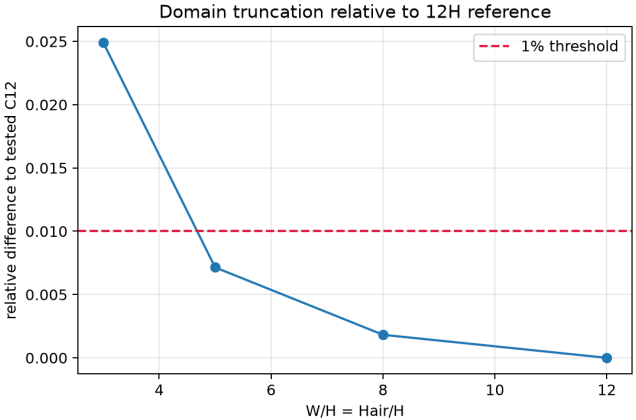
Smooth MMS

Observed orders: $L^2 \approx 2$, energy ≈ 1.

Takeaway

A small residual cannot by itself verify the material interface.

Domain truncation and the 8H choice



Relative to tested 12H reference

Domain	Difference
3H	2.491%
5H	0.714%
8H	0.182%
12H	reference

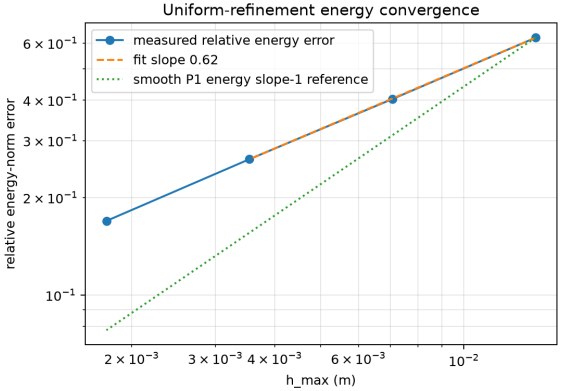
Mesh-adequacy repeat

Finer 12H repeat: 0.0360%. 8H reduces the tested difference versus 5H (0.182% vs 0.714%).

Takeaway

The 0.182% is a matched-core production sensitivity, not a strict M7 truncation-error bound.

Corner singularity and uniform convergence



Observed energy rate

Coarse fit M0–M2: $p_E \approx 0.623$ using M6 reference; ≈ 0.638 using M5.

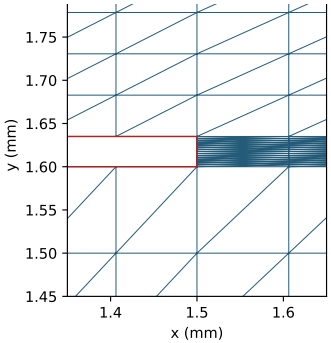
Interpretation

Corner and interface fields reduce regularity. The fitted rate depends on the reference and window.

Takeaway

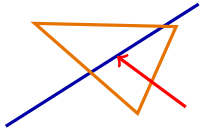
This is an observed coarse-mesh rate, not an exact singular exponent.

Why the Original Comparison Needed Correction



G2 corner mesh; M2: 404 DOFs, G2: 1,422 DOFs.

Geometry illustration only; costs differ.



Reference triangle crosses
a coarse-element edge

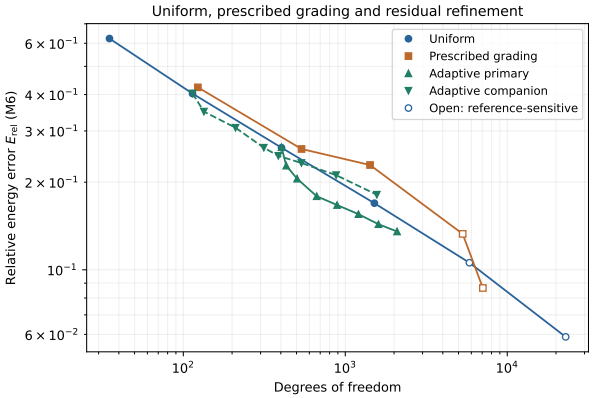
Schematic: blue = coarse edge; orange = reference triangle.

Fix: integrate on the triangle overlay.

Takeaway

Coordinate nesting does not ensure triangle nesting; the overlay resolves crossings.

Residual-driven adaptive refinement



Empirical error-versus-DOF slopes: -0.362 overall; -0.247 later.

At $E_{\mathrm{rel}} = 0.30$

$\sim 18\%$ **fewer DOFs**

Interpolated: adaptive 224 vs uniform 273 (M0-S).

Original cumulative time

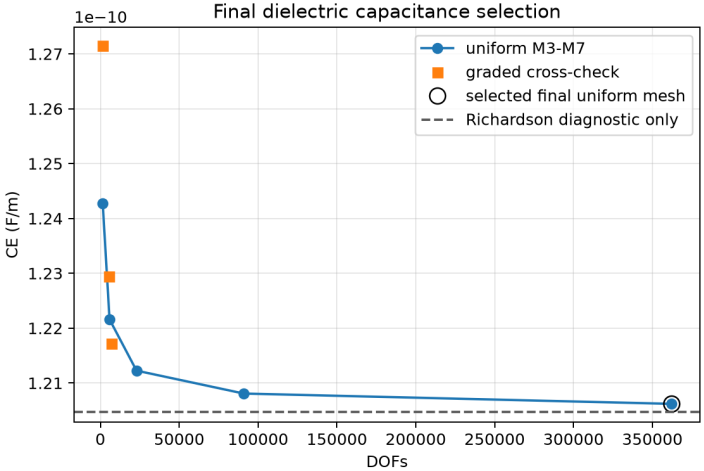
$1.83\times$ **uniform**

Includes preceding adaptive cycles.

Takeaway

Ideal $N^{-1/2}$ efficiency was not demonstrated.

Final capacitance selection



Selected result

Uniform M7, 8H domain

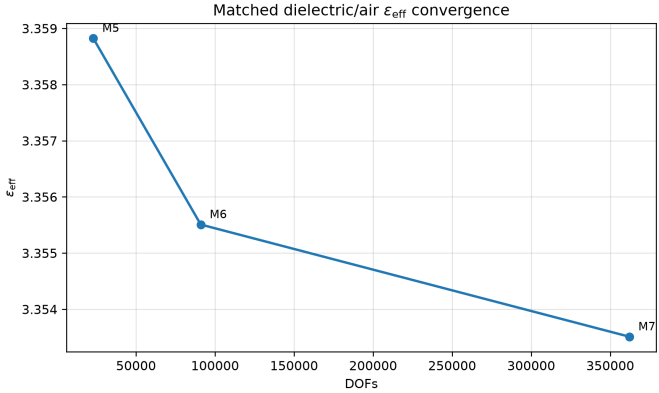
$$C = 120.620\,\mathrm{pF/m}$$

M6→M7 change: 0.1556%
Preferred QC target: 0.1% (not reached)

Why M7?

- Monotone capacitance convergence
- Stable asymptotic trend
- M8 rejected after resource preflight
- Richardson extrapolation is diagnostic only

Matched all-air solve and effective permittivity



Same frozen mesh

The all-air solve reuses the exact M7 mesh from the final dielectric case.

Matched capacitance ratio

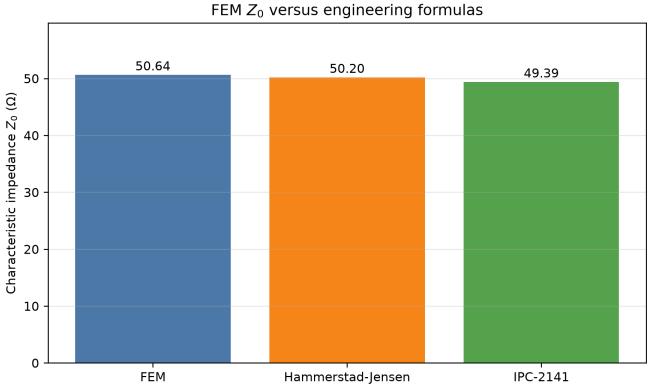
$$C_{\text{air}} = 35.9682\,\text{pF/m}$$

$$\varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}} = 3.3535$$

Takeaway

Mesh identity protects the ratio C/C_{air} from a hidden remeshing error.

Final Z_0 and engineering benchmarks



Quasi-TEM result

$$Z_0 = \frac{1}{c_0\sqrt{CC_{\text{air}}}} = 50.642\,\Omega$$

HJ-denominator discrepancy

Method	Z_0 (Ω)	$100d_{\text{HJ}}$
FEM	50.642	0.88%
HJ (finite t)	50.201	0
IPC-2141	49.393	1.61%

$$d_{\text{HJ}}(Z) = \frac{|Z - Z_{0,\text{HJ}}|}{|Z_{0,\text{HJ}}|}$$

Takeaway

Approximate benchmarks, not exact solutions. Sources: Hammerstad–Jensen (1980); IPC-2141 (1996).

Conclusions and limitations

Evidence and result

- Smooth and layered analytical tests check the solver.
- Triangle overlay corrected the cross-mesh comparison.
- $C = 120.620\,\mathrm{pF/m}$; $C_{\mathrm{air}} = 35.9682\,\mathrm{pF/m}$.
- $\varepsilon_{\mathrm{eff}} = 3.3535$; $Z_0 = 50.642\,\Omega$.

Limits

- Adaptive refinement saved DOFs, not cumulative time; ideal rate unproven.
- M7 and finite-domain sensitivities remain.
- Recorded Apple M2 run: 58 tests, 208 numerical comparisons passed; cross-platform install unverified.
- Quasi-static model omits losses and radiation.

Takeaway

A numerical result is credible when its verification and limitations are both visible.